

Negative-Energy Modalities in Laboratory Quantum Field Systems: Boundary, State, Tensor, and Readout Claims

Daniel S. Goldman
Independent researcher
ORCID: 0000-0003-2835-3521

June 1, 2026

Abstract

Laboratory discussions of negative energy often join several distinct physical objects under one phrase. A Casimir boundary condition, a prepared nonclassical field state, a renormalized stress-tensor component, an averaged energy-condition quantity, and an instrument response can each carry negative-energy language while licensing different conclusions. This paper sets out a compact claim grammar for those cases. The grammar separates four modalities: boundary-conditioned response, state-conditioned response, tensor/averaging relevance, and readout-conditioned observability.

The boundary-conditioned modality concerns vacuum response organized by material or geometric constraints, as in Casimir systems and finite worldline-numeric geometries. The state-conditioned modality concerns negative stress-energy created or selected by field-state preparation, measurement, squeezing, or quantum-energy-teleportation-type protocols. The tensor/averaging modality concerns the stress-energy object itself: which component is being discussed, how it is renormalized or smeared, which averaged condition it is compared against, and what source role it can play. The readout-conditioned modality concerns the apparatus channel: force, pressure, force gradient, resonant frequency, phase, loss, current, voltage, or detector response, together with the ordinary material and electromagnetic controls that define the measurement.

This vocabulary turns negative-energy claims into transferable but well-labeled statements. A boundary morphology can motivate a readout schedule; a state-preparation protocol can supply a local stress-energy deficit; a tensor calculation can compare the result with energy-condition or source requirements; a precision instrument can test whether the corresponding coefficient appears in measured data. The White et al. sphere-cylinder Casimir result is used as a worked boundary-conditioned example: its shell-like morphology, source-scale accounting, and physical readout routes occupy different modalities that can be evaluated together while preserving their meanings. The resulting framework is intended as a central reference for a paper pair: one companion audit assigns the White geometry to the boundary and tensor/source layers, and one companion readout paper asks how the shell-coded object could enter pressure, cavity, and superconducting measurement ensembles.

1. PURPOSE

Negative energy is a real and useful term in quantum field theory. It is also a term with several laboratory meanings. The Casimir effect supplies boundary conditioned vacuum response. Squeezed and measurement-conditioned states supply state-dependent local stress-energy structures. Quantum energy inequalities and averaged energy conditions supply duration, smearing, and source constraints.

Precision instruments supply force, pressure, frequency, phase, loss, current, voltage, and detector observables. Each of those statements belongs to a different physical modality.

The purpose of this paper is to make those modalities explicit. The goal is a positive claim grammar: a way to say exactly what has been shown, what kind of object carries the sign, which instrument can test it, and which additional evidence promotes it into a stronger claim. This is especially useful for finite Casimir geometries and exotic-spacetime-adjacent discussions, where a boundary-generated negative distribution can be visually suggestive while stress-tensor scale, averaged constraints, and apparatus readout remain separate questions.

The motivating case is the White, Vera, Han, Brucoleri, and MacArthur sphere–cylinder geometry [1]. In that work, worldline numerics applied to custom Casimir cavities produced a shell-like negative energy density distribution that was compared with the Alcubierre metric source shape. That comparison intersects several mature parts of the field: worldline methods for Casimir geometries [2], energy conditions and quantum energy inequalities [3, 4], precision Casimir metrology [9, 8], and emerging superconducting Casimir sensing [12]. The value of the example is that it asks all modality questions at once.

The central lesson is simple. A laboratory negative-energy claim becomes clear when it identifies the modality first. Boundary-conditioned statements describe how geometry organizes a vacuum response. State-conditioned statements describe how preparation or measurement changes a field state. Tensor/averaging statements describe the stress-energy object and its constraints. Readout-conditioned statements describe what the apparatus can recover against ordinary backgrounds. Those statements can support each other, and their interfaces are where the strongest experimental programs live.

2. FOUR MODALITIES

A modality is the physical route by which a negative-energy statement obtains meaning. It specifies the preparation or constraint, the stress-energy object, the spatial and temporal support, the compensation structure, the readout channel, and the inference licensed by the data. Table 1 gives the working classification.

The same experiment can occupy several modalities. A structured Casimir cavity can be boundary-conditioned in its source of vacuum response, tensor/averaging conditioned in its comparison with $T_{\mu\nu}$, and readout-conditioned in the force or resonator signal used to measure it. A quantum-energy-teleportation protocol can be state-conditioned in its preparation, tensor/averaging conditioned in its local deficit, and readout-conditioned in the detector or work-extraction channel used to score the event [5].

The classification is therefore an interface map rather than a taxonomy wall. It helps a paper state which object is being advanced and which object is being reserved for a companion calculation or experiment.

3. BOUNDARY-CONDITIONED RESPONSE

The boundary-conditioned modality begins with a material or geometric constraint on quantum fields. Parallel plates, sphere–plate systems, microstructured surfaces, deep trenches, nanomembranes, and superconducting microplates all fit this class. The signed object is the boundary-conditioned change in vacuum energy, stress, pressure, force, or a calculational proxy for those quantities. The physical control is geometry and material response.

This modality is the native home of Casimir physics. Precision measurements typically recover force, force gradient, pressure, or resonator response, with surface potentials, roughness, edge effects,

Modality	Primary object	Licensed statement
Boundary-conditioned	Vacuum response fixed by material boundaries, geometry, and field boundary conditions	A finite apparatus geometry organizes Casimir energy, pressure, stress, or a proxy density into a spatial pattern.
State-conditioned	Stress-energy in a prepared, post-selected, squeezed, measured, or feed-forward-conditioned quantum field state	A field state carries a local or smeared stress-energy deficit with a defined preparation record and compensation structure.
Tensor/averaging	Renormalized stress tensor, null or timelike contraction, smeared integral, or energy-condition quantity	The signed object is tied to a specified component, averaging window, source comparison, and quantum-inequality or averaged-condition context.
Readout-conditioned	Measured apparatus response: force, pressure, gradient, frequency, phase, loss, current, voltage, transition, or detector signal	A laboratory observable contains a recoverable coefficient with declared controls, nuisance templates, calibration, and false-positive logic.

Table 1: Four laboratory modalities for negative-energy claims. The modalities can be combined, but each licenses a different kind of conclusion.

finite conductivity, temperature, alignment, and material optical data treated as part of the measurement model [9, 10, 11]. MEMS/NEMS and nanomembrane platforms make the same point experimentally: the observable is an apparatus response whose interpretation depends on careful material and electrostatic accounting [6, 7, 8].

Worldline numerics adds a valuable finite-geometry language to this modality. It can evaluate Casimir energies and, in stress-tensor extensions, components of the fluctuation-induced energy-momentum tensor in arbitrary geometries [2]. That makes it natural to ask shape questions: where the response concentrates, how it shifts under geometry perturbations, how much of the signed weight lies in a chosen region, and whether the strongest response tracks a designed boundary feature.

The White sphere–cylinder example is boundary-conditioned at this level. A $1\text{ }\mu\text{m}$ diameter sphere centered in a nominal $4\text{ }\mu\text{m}$ diameter cylinder produces a shell-like distribution in the reported worldline analysis [1]. The field-facing boundary question is whether a finite sphere–wall geometry organizes the vacuum response into an annular shell role. That question is independent from the larger source-scale and direct readout questions. It asks what the boundary does.

4. STATE-CONDITIONED RESPONSE

The state-conditioned modality begins with a quantum field state rather than a static boundary geometry. The signed object is a local or smeared stress-energy feature of that state. Examples include squeezed-state stress-energy profiles, measurement-conditioned field states, and quantum-energy-teleportation-type protocols in which local operations and classical records create a negative stress-energy density in a target region [3, 5].

The preparation record is part of the claim. It tells the reader how the field state was made, which degrees of freedom were measured or driven, how the classical record enters, where compensating positive energy appears, and what time window carries the deficit. This modality is dynamic and

operational. Its strength is that the sign can be tied to a controllable state-preparation procedure alongside fixed-boundary inference.

State-conditioned negative energy also naturally meets quantum energy inequalities. Ford’s review emphasizes the basic QFT fact that local negative energy densities occur, with magnitude and duration governed by quantum constraints [3]. Kontou and Sanders place that fact inside the broader energy condition landscape: quantum fields motivate a refined move from pointwise conditions to averaged and quantum-inequality statements [4]. For state-conditioned claims, duration, smearing, compensation, and averaging are part of the observable content.

This modality is useful beside boundary-conditioned Casimir work because it clarifies an important distinction. A Casimir geometry changes the vacuum state through boundary conditions. A QET-style or squeezed-state protocol changes, selects, or conditions the field state through operations. Both can produce negative stress-energy language. Their preparation logic, compensation logic, and readout logic are different.

5. TENSOR AND AVERAGING RELEVANCE

The tensor/averaging modality asks which stress-energy object is being used. For a relativistic source claim, the relevant object is more than a visual negative region. It is a component or contraction of the renormalized stress tensor, such as T_{00} , $T_{ab}k^ak^b$, or an integral of T_{ab} over a specified sampling domain. The tensor object then has to be compared with the source role under discussion.

This modality supplies the bridge from laboratory QFT to gravity-facing language. Energy conditions in general relativity were historically formulated as restrictions on stress-energy; quantum field theory gives well-understood violations of pointwise versions and motivates averaged or quantum-inequality replacements [4]. A laboratory negative-energy measurement can therefore be precise about its level:

- local energy density or pressure in a material/boundary setup;
- a stress-tensor component or finite-difference proxy;
- a null or timelike contraction;
- a smeared or averaged quantity;
- a source-demand comparison for a specified metric model.

This layer is where morphology and source relevance separate cleanly. A boundary-conditioned morphology can be real and reproducible while its calibrated stress scale remains Casimir-scale relative to a gravity-facing source demand. A state-conditioned deficit can be operationally generated while its averaged integrals obey quantum-inequality restrictions. A readout can recover a geometry-locked coefficient while the corresponding tensor component remains an ordinary Casimir-scale object.

The positive role of the tensor/averaging modality is to keep each of those statements useful. It lets morphology remain morphology, readout remain readout, and source comparison remain source comparison. It gives the field a way to promote claims by adding tensor and averaging evidence while preserving the natural conclusion of each modality.

6. READOUT-CONDITIONED OBSERVABILITY

The readout-conditioned modality begins with an instrument. Casimir experiments measure force, force gradient, pressure, displacement, oscillator-frequency shift, phase, quality factor, transition behavior, or other apparatus responses. Those responses are sensitive to vacuum-boundary physics through a transduction chain, and the same chain also carries ordinary material and electromagnetic effects.

This modality is where laboratory claims become data statements. A readout claim should identify:

- the observable and units;

- the geometry schedule or state-preparation schedule;
- calibration channels and null controls;
- material, patch, thermal, vibration, alignment, and readout-chain templates;
- the recovery statistic and false-positive test.

Modern Casimir metrology already works in this style. Bimonte et al. combine micromechanical torsional-oscillator measurements with explicit roughness, edge, patch-potential, and optical-data modeling over $0.2\text{--}8\,\mu\text{m}$ separations [9]. Patch-potential modeling and Kelvin-probe measurements are central to high-quality force interpretation [10, 11]. Superconducting-transition Casimir sensing adds a cryogenic readout frontier in which tiny pressure changes are compared against electrostatic, thermal, material, and alignment effects [12].

Readout-conditioned observability is therefore a positive experimental language. It gives a boundary or state result an instrument-native test. It also keeps the measured object specific: pressure, frequency shift, phase, loss, current, and voltage each carry a distinct transfer function, nuisance structure, and claim boundary.

7. TRANSFERS BETWEEN MODALITIES

Most strong laboratory papers move across modalities. A boundary calculation motivates an instrument. A state-preparation protocol predicts a stress-tensor deficit. A readout experiment recovers a coefficient that can be compared with a tensor model. The transfer is scientifically productive when the promoted claim carries the extra evidence required by the new modality.

Table 2 lists common transfers and their evidence requirements.

Transfer	Added evidence	Promoted conclusion
Boundary morphology to tensor object	Stress-tensor component, normalization, convergence, material model, and smearing convention	The shape is associated with a quantified energy, pressure, or stress channel.
Tensor object to source comparison	Metric source demand, units, sign convention, averaging window, and scale ratio	The laboratory stress-energy object can be compared with a declared gravity-facing source role.
Boundary or state object to readout	Observable model, transfer function, controls, nuisance templates, and false-positive test	The apparatus can search for the corresponding coefficient in measured data.
Readout coefficient to field interpretation	Calibration closure, background separation, phase/geometry locking, and replication under controls	The measured signal can be assigned to the boundary or state-conditioned field object under the declared model.
State preparation to averaged-energy statement	Sampling function, duration, compensating positive energy, and quantum inequality comparison	The local deficit is placed inside a finite-time and finite-region energy ledger.

Table 2: Evidence required when a claim moves from one modality to another. The promoted conclusion names the new physical object rather than relabeling the old one.

This transfer table is the practical core of the modality grammar. It protects useful intermediate results. A morphology result can stand as a morphology result. A readout architecture can stand as a readout architecture. A source comparison can stand as a source comparison. The strongest papers then show how those results connect.

8. WHITE SPHERE–CYLINDER CASE STUDY

The White sphere–cylinder result is an ideal case study because it touches all four modalities. The original paper reports a boundary-conditioned worldline-numerics result: a custom Casimir geometry produces a shell-like negative distribution in a toy model consisting of a $1\,\mu\text{m}$ sphere centered in a $4\,\mu\text{m}$ cylinder [1]. The visual resemblance to an Alcubierre-style source shape makes the tensor/source modality relevant. The proposed voltage, current, photon, or electron detection ideas make the readout modality relevant. The public interpretation then depends on keeping those modalities aligned.

The companion audit in this paper ensemble treats the White geometry as a boundary-conditioned and tensor/source case. Its role-resolved question is: what remains after separating the sphere, cylinder wall, annular shell, central readout path, far field, material/background accounting, and source scale? Within that structure, the shell-like boundary morphology can be supported while direct gravity-facing scale and central timing/readout routes are bounded by calibration and recovery gates.

The companion readout paper treats the shell-coded object as a readout-conditioned target. Its question is: which physical apparatus ensembles could test a geometry-locked boundary response? Pressure cells, high- Q cavities, and superconducting resonators become branch-native readouts. Their success criteria are pressure contrast, eigenmode shift, phase/loss response, or cryogenic microwave response with material and electromagnetic controls.

Placed in the modality grammar, the case study has a clean identity:

- **Boundary modality:** the finite sphere–cylinder geometry organizes a shell-like Casimir response.
- **Tensor/averaging modality:** stress scale, source demand, and averaging decide the gravity-facing interpretation.
- **Readout modality:** pressure, cavity, superconducting, voltage, current, photon, or electron channels decide what an apparatus can recover.
- **State modality:** the White setup is primarily boundary-conditioned; any state-conditioned extension would require an additional preparation protocol.

This is the value of the central paper. It lets the White pair stay technically sharp while the ensemble has a shared conceptual grammar. It also lets other negative-energy experiments locate themselves on the same map.

9. FIELD-FACING CLAIM GRAMMAR

The field-facing version of the framework is a short checklist. A laboratory negative-energy paper can state:

1. **Modality.** Is the sign boundary-conditioned, state-conditioned, tensor/averaging-conditioned, readout-conditioned, or a declared combination?
2. **Object.** Is the object energy, pressure, force, stress, frequency shift, phase, loss, current, voltage, detector response, or a model coefficient?
3. **Support.** Where and when does the signed object live, and how is the support defined?
4. **Compensation.** Where does the associated positive energy, material response, reset cost, or background response appear?
5. **Controls.** Which material, electromagnetic, thermal, patch, alignment, drift, and readout-chain controls share the same apparatus?
6. **Promotion path.** Which additional evidence would move the result into a stronger modality?

Table 3 summarizes the resulting claim levels.

This grammar also improves peer review. A reviewer can ask whether the paper’s evidence matches the modality being claimed. The authors can then answer at the right level: boundary morphology, field-state preparation, stress-tensor scale, readout recovery, or source relevance.

Claim level	Evidence basis	Clean conclusion
Geometry organizes response	Finite-boundary calculation, perturbation checks, morphology metrics	The apparatus geometry carries a reproducible boundary-conditioned pattern.
Stress channel quantified	Stress tensor or calibrated proxy, material model, scale estimate	The pattern has a quantified energy, pressure, or stress role.
Operational state deficit	Preparation record, detector/work ledger, duration and compensation	The field state carries a controlled local or smeared deficit.
Instrument recovers coefficient	Measured observable, controls, template separation, false-positive test	The apparatus sees a recoverable boundary or state-conditioned coefficient.
Source relevance established	Tensor component, averaging, source-demand comparison, scale closure	The stress-energy object can be assigned to the declared gravity-facing source role.

Table 3: Claim levels enabled by the modality grammar. Each level is useful on its own and can be promoted by adding the evidence named in the transfer table.

10. IMPLICATIONS FOR LABORATORY PROGRAMS

The modality grammar suggests a constructive path for laboratory QFT programs. First, identify the modality with the strongest existing evidence. For many Casimir platforms, that is boundary-conditioned response plus readout-conditioned force or pressure metrology. For QET-style protocols, it is state-conditioned preparation plus operational energy extraction or detector response. For energy-condition studies, it is tensor/averaging structure plus duration and smearing constraints.

Second, design the transfer deliberately. A finite-boundary morphology needs a stress or readout model before it becomes an apparatus claim. A readout signal needs template separation before it becomes a field claim. A local stress-energy deficit needs smearing and compensation before it becomes an averaged-energy statement. A gravity-facing source comparison needs calibrated stress scale, tensor component, sign convention, and source-demand ratio.

Third, treat ordinary apparatus physics as part of the field experiment. Material response, patch potentials, finite conductivity, thermal behavior, roughness, mode mixing, vibration, alignment, and readout-chain effects are the channels through which laboratory negative-energy claims become precise. In high-quality Casimir work, those channels already provide the calibration structure that makes the measurement scientific [9, 10, 11, 12].

This has a positive consequence for speculative-adjacent systems. A visually suggestive negative-energy morphology can be valuable at an early source-assignment stage. It can be a boundary-conditioned target, a stress-channel hypothesis, a readout template, or a discrimination schedule. Each role supports real progress when it is named.

11. REPRODUCIBILITY AND SOURCE BASIS

The cited source PDFs for this central paper are mirrored in the local `white_casimir_intersection/sources` directory. That directory includes the White et al. EPJ C paper, QFT negative-energy and energy-condition references, the worldline stress-tensor reference, engineered negative stress-energy via quantum energy teleportation, and the Casimir metrology, patch-potential, and superconducting-transition references used for the readout discussion.

This manuscript is a conceptual synthesis. Its reproducible content is the claim grammar: the modality definitions, transfer-evidence table, and claim-level table. The two companion papers supply the worked White sphere–cylinder audit and the physical readout architecture application.

12. CONCLUSION

Negative-energy language is most powerful when its modality is explicit. Boundary-conditioned response, state-conditioned response, tensor/averaging relevance, and readout-conditioned observability are different laboratory objects. They can reinforce one another, and their interfaces define strong experimental programs.

The White sphere–cylinder case illustrates the payoff. The shell-like Casimir boundary morphology, the stress/source scale, and the proposed readout routes are connected but distinct. The central paper supplies the vocabulary for that separation. The companion audit applies it to the White geometry. The companion readout paper applies it to pressure, cavity, and superconducting measurement ensembles.

The field-level result is a clean claim grammar for laboratory QFT. A paper can now say which kind of negative energy it has, which object carries the sign, which controls make the readout meaningful, and which additional evidence would promote the result into a stronger modality. That grammar keeps boundary effects, prepared field states, stress-tensor constraints, and instrument responses scientifically connected while preserving the specific meaning of each.

References

- [1] H. White, J. Vera, A. Han, A. R. Brucoleri, and J. MacArthur, “Worldline numerics applied to custom Casimir geometry generates unanticipated intersection with Alcubierre warp metric,” *European Physical Journal C* **81**, 677 (2021).
<https://doi.org/10.1140/epjc/s10052-021-09484-z>.
- [2] M. Schafer, I. Huet, and H. Gies, “Worldline Numerics for Energy-Momentum Tensors in Casimir Geometries,” arXiv:1509.03509 (2015). <https://arxiv.org/abs/1509.03509>.
- [3] L. H. Ford, “Negative Energy Densities in Quantum Field Theory,” arXiv:0911.3597 (2009).
<https://arxiv.org/abs/0911.3597>.
- [4] E.-A. Kontou and K. Sanders, “Energy conditions in general relativity and quantum field theory,” *Classical and Quantum Gravity* **37**, 193001 (2020).
<https://doi.org/10.1088/1361-6382/ab8fcf>.
- [5] N. Funai and E. Martin-Martinez, “Engineering negative stress-energy densities with quantum energy teleportation,” *Physical Review D* **96**, 025014 (2017).
<https://doi.org/10.1103/PhysRevD.96.025014>.
- [6] R. S. Decca, V. Aksyuk, and D. Lopez, “Casimir force in micro and nano electro mechanical systems,” in *Casimir Physics*, Lecture Notes in Physics **834** (2011).
https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=906575.
- [7] D. Garcia-Sanchez, K. Y. Fong, H. Bhaskaran, S. Lamoreaux, and H. X. Tang, “Casimir Force and In Situ Surface Potential Measurements on Nanomembranes,” *Physical Review Letters* **109**, 027202 (2012). <https://doi.org/10.1103/PhysRevLett.109.027202>.

- [8] A. Stange, M. Imboden, J. Javor, L. K. Barrett, and D. J. Bishop, “Building a Casimir metrology platform with a commercial MEMS sensor,” *Microsystems & Nanoengineering* **5**, 14 (2019). <https://doi.org/10.1038/s41378-019-0054-5>.
- [9] G. Bimonte, B. Spreng, P. A. Maia Neto, G.-L. Ingold, G. L. Klimchitskaya, V. M. Mostepanenko, and R. S. Decca, “Measurement of the Casimir Force between 0.2 and 8 μm : Experimental Procedures and Comparison with Theory,” *Universe* **7**, 93 (2021). <https://doi.org/10.3390/universe7040093>.
- [10] R. O. Behunin, F. Intravaia, D. A. R. Dalvit, P. A. Maia Neto, and S. Reynaud, “Modeling electrostatic patch effects in Casimir force measurements,” arXiv:1108.1761 (2011). <https://arxiv.org/abs/1108.1761>.
- [11] J. L. Garrett, D. Somers, and J. N. Munday, “The effect of patch potentials in Casimir force measurements determined by heterodyne Kelvin probe force microscopy,” *Journal of Physics: Condensed Matter* **27**, 214012 (2015). <https://doi.org/10.1088/0953-8984/27/21/214012>.
- [12] M. Xu, R. J. G. Elbertse, A. Keskekler, G. Bimonte, J. Lee, S. Otte, and R. A. Norte, “High-Resolution Casimir Force Sensing Across a Superconducting Transition,” arXiv:2504.10579, version 2 (2026). <https://arxiv.org/abs/2504.10579>.